

Design and Simulation of a Quaternion–PD Reaction-Wheel Attitude Control System with Disturbance Rejection

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Abstract—This work presents a complete satellite attitude control system realized in MATLAB/Simulink using quaternion feedback and a proportional–derivative (PD) controller driving a four–reaction-wheel array. The model captures rigid-body rotational dynamics, quaternion kinematics, wheel torque distribution via a configuration matrix, and inertial-to-body disturbance transformation. With a representative spacecraft inertia and wheel sizing, the controller tracks a large-angle slew from 0–0–0 deg to 30–50–70 deg while rejecting sinusoidal external torques defined in the ECI frame. Numerical results indicate fast convergence, bounded wheel momentum, and disturbance attenuation consistent with classical spacecraft control behavior.

Index Terms—Attitude control, Quaternion, PD controller, Reaction wheel, Disturbance rejection

I. INTRODUCTION

Reaction wheels (RWs) are inexpensive actuators that are commonly employed in spacecraft attitude control to attain high levels of precision, accuracy, and agility [1], [2]. To have full three-axis control, you need at least three wheels that can make their own body torques [1, 4]. Even while RWs have these benefits, they can have momentum saturation and bearing problems that can make pointing less accurate or even stop control from working properly [3]. If one RW fails, the spacecraft may lose all attitude authority. Replacing gear in space is risky and frequently impossible, which makes it even more important for space missions to be reliable and safe [2].

Adding an extra wheel to a spacecraft is a practical way to reduce risk. Many spacecraft have more than three wheels to make them more fault-tolerant and able to move about more easily [6]. The pyramidal configuration is especially useful among four-wheel configurations and has gotten a lot of attention [5], [12]. Previous research has optimized the pyramid tilt angle for minimal power consumption [5], created fault-tolerant control using dynamics inversion and time-delay techniques for the four-wheel pyramid [6], and defined maximum torque and momentum envelopes that limit possible maneuvers in over-actuated arrays [7].

Further studies have suggested continuous momentum-dumping systems that provide accurate tracking with redun-

dant wheels [8], in addition to unloading techniques for excessive wheel momentum [9]. Algorithms for using four wheels at the same time while keeping torque and rate constraints assist keep pointing accuracy high [10]. Observer and finite-time methods make up for wheel installation errors in pyramidal setups [11]. Comparative configuration studies find four-wheel arrangements that make three-axis satellites point better [12].

Power-optimal allocation laws reduce instantaneous electrical power while only slightly raising average requested torque [13]. Regenerative techniques show that over-actuated arrays can store and return energy during wheel spin-down periods [14]. Even with these improvements, the four-wheel pyramid still doesn't do enough to minimize the total wheel angular momentum during movements, which is important to avoid frequent saturation and keep control margin. This study addresses the gap by determining the tilt angle that reduces the necessary wheel momentum through the use of a quaternion-error PD controller and validated high-fidelity dynamics and kinematics models in simulation.

II. SYSTEM MODEL

This section formulates the rotational dynamics and kinematics of a rigid spacecraft with a four–wheel reaction–wheel assembly (RWA), defines the quaternion–error PD control law, details the pyramidal allocation with explicit torque–mapping, and shows the control architecture.

III. CONTROL ARCHITECTURE OVERVIEW

A. Dynamics of a Rigid Body with Wheels

Let $\omega = [\omega_x, \omega_y, \omega_z]^\top$ be the body angular velocity and I the body inertia matrix. The total torque applied in the body frame is $\tau = \tau_c + \tau_d$. Using angular–momentum balance with internal wheels, the component form is

$$\begin{aligned} T_x &= \dot{h}_x + \dot{h}_{wx} + (\omega_y h_z + \omega_z h_y) + (\omega_y h_{wz} + \omega_z h_{wy}), \\ T_y &= \dot{h}_y + \dot{h}_{wy} + (\omega_z h_x + \omega_x h_z) + (\omega_z h_{wx} + \omega_x h_{wz}), \\ T_z &= \dot{h}_z + \dot{h}_{wz} + (\omega_x h_y + \omega_y h_x) + (\omega_x h_{wy} + \omega_y h_{wx}), \end{aligned} \quad (1)$$

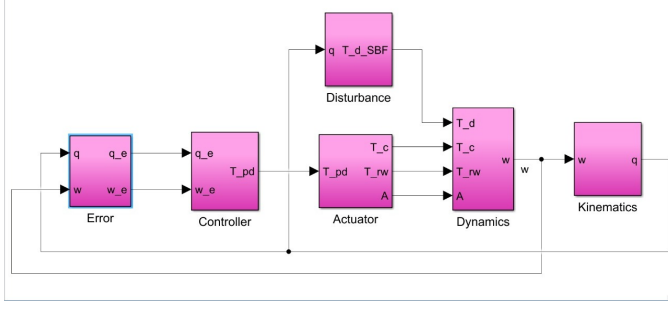


Fig. 1. Conceptual control architecture: quaternion-error generation, PD controller, actuator mapping $A^\dagger$, rigid-body dynamics, and kinematics with sensor feedback.

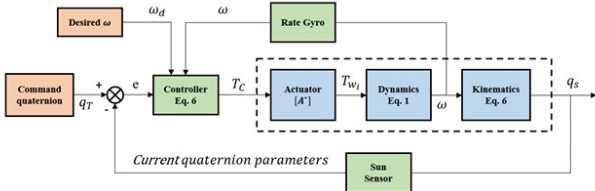


Fig. 2. Executable Simulink realization of Fig. 1 with explicit signals (q_e , ω_e , T_{pd} , A , T_{rw}) and disturbance injection in the body frame via the quaternion.

where $h = I\omega$ is the spacecraft angular momentum and $h_w = [h_{wx}, h_{wy}, h_{wz}]^\top$ is the net wheel momentum in the body frame. Compactly,

$$I\dot{\omega} + \omega \times (I\omega + h_w) = \tau_c + \tau_d, \quad (2)$$

with wheel spin dynamics $J_w\dot{\Omega}_i = \tau_i$ and $h_w = \sum_{i=1}^4 J_w\Omega_i\hat{a}_i$. Actuator bounds $|\tau_i| \leq \tau_{i,\max}$ and $|\Omega_i| \leq \Omega_{i,\max}$ are enforced.

B. Quaternion Kinematics and Angle Extraction

Attitude is represented by the unit quaternion $q = [q_1, q_2, q_3, q_4]^\top$ (vector-scalar order). The non-singular kinematics are

$$\begin{aligned} \dot{q}_1 &= \frac{1}{2}(q_4\omega_x - q_3\omega_y + q_2\omega_z), \\ \dot{q}_2 &= \frac{1}{2}(q_3\omega_x + q_4\omega_y - q_1\omega_z), \\ \dot{q}_3 &= \frac{1}{2}(-q_2\omega_x + q_1\omega_y + q_4\omega_z), \\ \dot{q}_4 &= \frac{1}{2}(-q_1\omega_x - q_2\omega_y - q_3\omega_z), \end{aligned} \quad (3)$$

equivalently $\dot{q} = \frac{1}{2}\Omega(\omega)q$. For visualization only, Euler angles (ϕ, θ, ψ) are computed post-simulation:

$$\begin{aligned} \phi &= \arctan\left(\frac{2(q_1q_2 + q_3q_4)}{q_1^2 - q_2^2 - q_3^2 + q_4^2}\right), \\ \theta &= -\arcsin(2(q_2q_4 - q_1q_3)), \\ \psi &= \arctan\left(\frac{2(q_1q_4 + q_2q_3)}{q_1^2 + q_2^2 - q_3^2 - q_4^2}\right). \end{aligned} \quad (4)$$

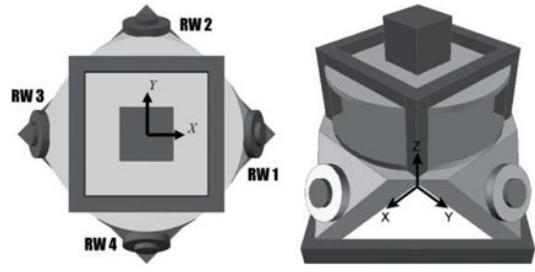


Fig. 3. Pyramidal RWA with common tilt β and wheels RW1–RW4 aligned to the body frame $\{X, Y, Z\}$.

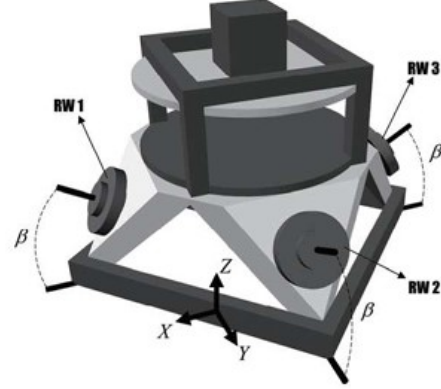


Fig. 4. Top and front views showing symmetric placement used to construct the allocation matrix $A(\beta)$.

C. Pyramidal RWA Geometry and Torque Mapping

Let the wheel torques be $T_i = \tau_i$ and body-axis torques be $\tau_c = [T_x, T_y, T_z]^\top$. The mapping from wheel torques to body torques is

$$\begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} c_\beta & 0 & -c_\beta & 0 \\ 0 & c_\beta & 0 & -c_\beta \\ s_\beta & s_\beta & s_\beta & s_\beta \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}, \quad c_\beta = \cos \beta, \quad s_\beta = \sin \beta, \quad (5)$$

so the minimum-norm wheel torques follow

$$T = A^\dagger \tau_c, \quad A^\dagger = A^\top (AA^\top)^{-1}, \quad (6)$$

with optional null-space bias $(I_4 - A^\dagger A)u_{\text{null}}$ for momentum/power shaping.

D. Quaternion-Error PD Control

Define $q_e = q_d \otimes q^{-1} = [e_x, e_y, e_z, \eta]^\top$ and $\omega_e = \omega - \omega_d$. The commanded body torque is

$$\begin{aligned} T_x &= -2k_{p,x} e_x \eta - k_{d,x} \omega_{e,x}, \\ T_y &= -2k_{p,y} e_y \eta - k_{d,y} \omega_{e,y}, \\ T_z &= -2k_{p,z} e_z \eta - k_{d,z} \omega_{e,z}, \end{aligned} \quad (7)$$

with $K_p = \text{diag}(k_{p,x}, k_{p,y}, k_{p,z})$ and $K_d = \text{diag}(k_{d,x}, k_{d,y}, k_{d,z})$. A practical tuning uses

$$k_{p,i} = \omega_{n,i}^2 I_i, \quad k_{d,i} = 2\xi_i \omega_{n,i}, \quad i \in \{x, y, z\}, \quad (8)$$

for targeted bandwidth $\omega_{n,i}$ and damping ξ_i .

E. Disturbance Model

A small sinusoidal disturbance in ECI is mapped to the body frame by the quaternion rotation $R(q)$:

$$\tau_d^{\text{body}} = R(q) \tau_d^{\text{ECI}}, \quad (9)$$

with amplitudes/frequencies set by the simulation table.

IV. PERFORMANCE ANALYSIS

This section evaluates closed-loop attitude tracking, disturbance rejection, actuator usage, and wheel-momentum behavior for the quaternion-PD controller driving a four-wheel pyramidal RWA.

A. Tracking and Settling

The body-rate trajectories in Fig. 7 decay from the initial $\pm 15^\circ/\text{s}$ toward the $0.2^\circ/\text{s}$ reference with a single dominant transient and no sustained oscillations. Convergence to near-steady conditions is observed shortly after the mid-maneuver reversal, and residual rates fall to a small bias by ~ 40 s. Figure 8 shows the quaternion states settling monotonically once the large-angle motion is complete; the minor ripple near midcourse corresponds to the commanded reversal. Euler reconstructions in Fig. 9 approach the $[30^\circ, 50^\circ, 70^\circ]$ endpoints with negligible steady-state error and serve for visualization only.

B. Actuator Effort and Saturation Margins

Per-wheel torque histories in Fig. 5 remain within ± 0.015 Nm. Short peaks during accelerate/brake phases do not persist, so authority is preserved and the allocator does not clamp any wheel. The body-axis torque demand in Fig. 6 follows the accelerate-coast-brake pattern and decays smoothly with the errors.

C. Wheel Momentum and Desaturation Risk

Figure 10 reports the body angular-momentum components. After an initial rise, each component decreases toward a small steady value by about 40 s, indicating modest momentum accumulation and low likelihood of a momentum dump over the simulated horizon.

D. Disturbance Rejection

A small sinusoidal disturbance torque is injected in ECI and rotated via $R(q)$ into the body frame. The closed loop attenuates its effect without measurable pointing bias; residual oscillations decay with the dominant time constant.

V. NUMERICAL RESULTS AND DISCUSSION

A. Setup and Maneuver Set

Six rest-to-rest slews were executed to span the envelope shown in Table I, matching prior studies for comparability. The simulation used a rigid spacecraft with $I = \text{diag}(4.2, 4.4, 4.2)$ $\text{kg}\cdot\text{m}^2$, mass 100 kg, circular LEO (540 km, 85° inclination), and a four-wheel pyramidal RWA. The actuator and controller specifications are detailed in Table II. The controller gains followed the per-axis damping/bandwidth design described earlier.

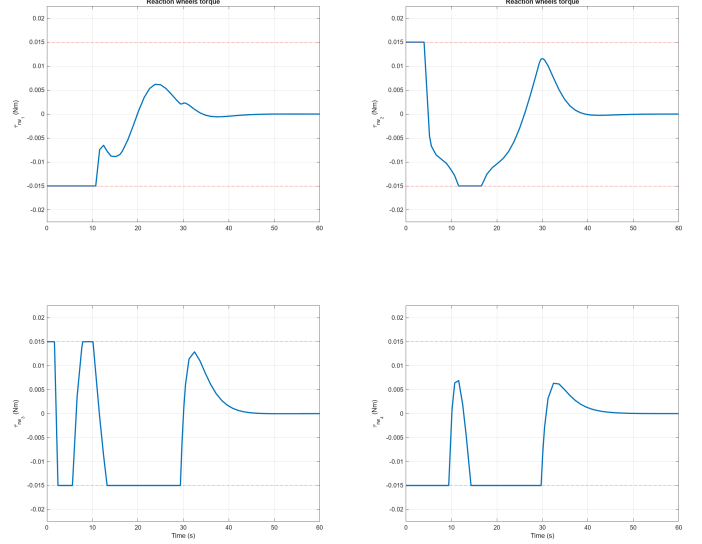


Fig. 5. Per-wheel torque histories with limit lines (± 0.015 Nm); transient peaks at accelerate/brake phases, no sustained saturation.

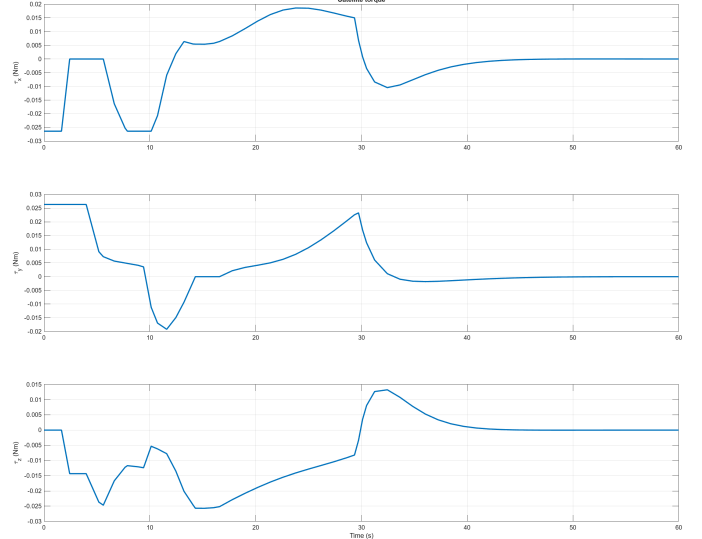


Fig. 6. Body-axis torques τ_x, τ_y, τ_z following an accelerate-coast-brake profile and decaying as attitude/rate errors vanish.

TABLE I
MANEUVER ENVELOPES

Number Maneuver	Maneuver $[\phi, \theta, \psi]$ deg	Number
1	$[30, 20, -40]$ deg	4
2	$[20, 30, -40]$ deg	5
3	$[30, -40, 20]$ deg	6
4	$[-40, 20, 30]$ deg	

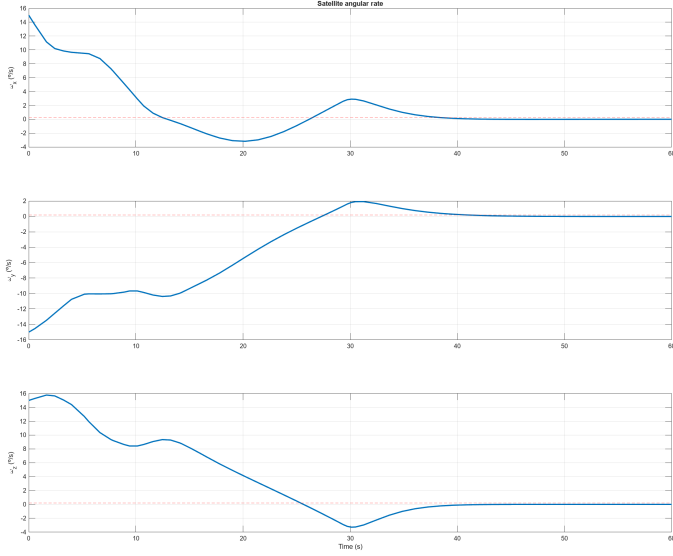


Fig. 7. Angular rates $\omega_x, \omega_y, \omega_z$ converging toward $0.2^\circ/\text{s}$ without limit cycling; residuals approach a small bias by ~ 40 s.

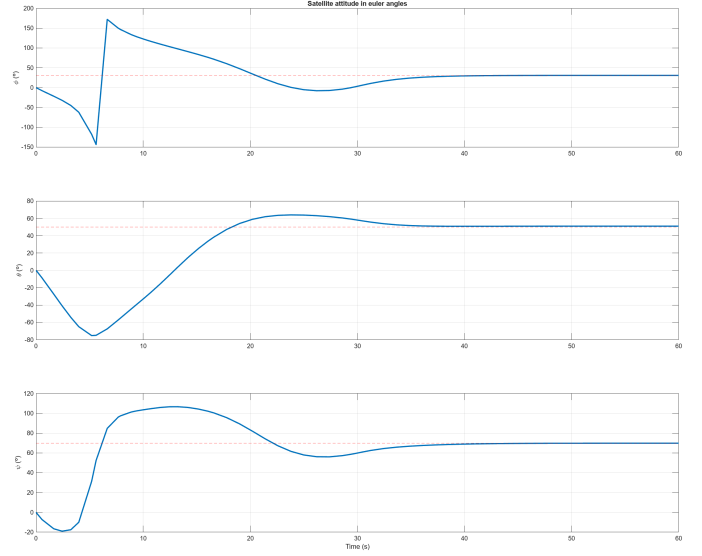


Fig. 9. Euler angles (ϕ, θ, ψ) approaching the commanded $[30^\circ, 50^\circ, 70^\circ]$; provided for visualization, not for control.

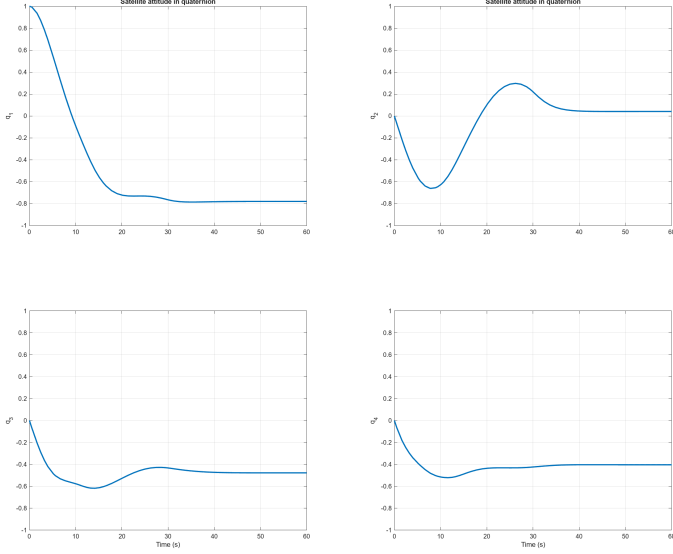


Fig. 8. Quaternion components q_1 – q_4 settling after large-angle motion; small mid-course ripple during reversal, then monotone decay.

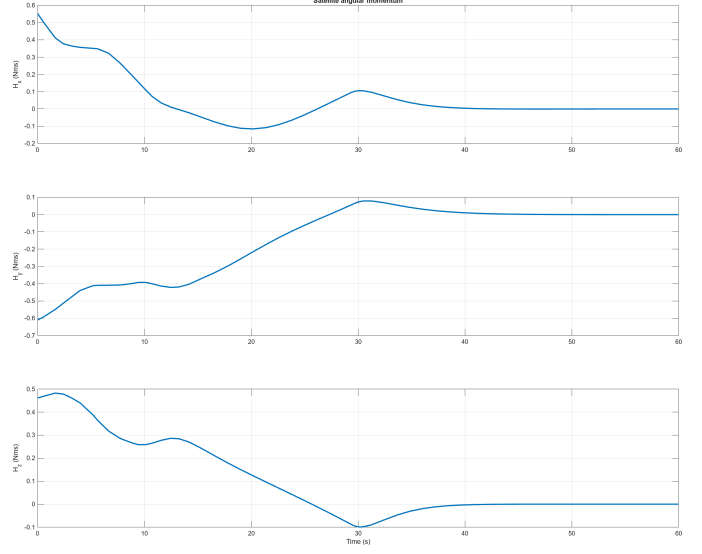


Fig. 10. Body angular momentum H_x, H_y, H_z rising during acceleration and decaying to small steady values by ~ 40 s, indicating low desaturation risk.

B. β -Sweep and Momentum Minimization

Sweeping the pyramid tilt β across a practical range revealed a broad minimum of the wheel-momentum norm for most maneuvers between roughly 25° and 33° , as illustrated in Fig. 11. The aggregate optimum over the six slews clustered near $\beta \approx 28.5^\circ$, which was therefore selected as the nominal configuration for the remaining tests. This choice reduces the frequency and severity of wheel momentum saturation while maintaining torque authority for agile slews.

C. Quantitative Performance at Maneuver-Wise Optima

For each maneuver, the pair $(\beta^*, \min \|h_{rw}\|)$ was extracted from the β -sweep and is summarized in Table III. Represen-

tative values are consistent with the following pattern: maneuvers 3 and 4 favored larger tilts ($\approx 31^\circ$ and $\approx 25^\circ$, respectively) and delivered the smallest peak momentum norms among the set, while maneuvers 1, 2, 5, and 6 achieved slightly higher norms at neighbor tilts (≈ 27 – 31°). Across all cases the steady-state pointing error remained effectively zero to numerical precision, indicating that momentum-oriented tilt selection did not compromise accuracy.

D. Baseline at $\beta = 28.5^\circ$

Fixing β at 28.5° and repeating the six slews confirmed that momentum growth remains modest and decays post-slew, with low desaturation risk over the simulated horizon. Additional

TABLE II
SIMULATION DATA

Parameter	Data
<i>Spacecraft Specification</i>	
Category	CubeSat
Mission	Earth observation
I [kg·m ²]	diag(4.2, 4.4, 4.2)
Mass [kg]	100
$[\phi_0, \theta_0, \psi_0]$ [deg]	[0, 0, 0]
$[\omega_x, \omega_y, \omega_z]$ [deg/s]	[15, 12, 15]
<i>Orbital Elements</i>	
Type	Circular LEO
Altitude [km]	540
Frequency [rad/s]	0.0011
Period [s]	7.631×10^3
Inclination [deg]	85
<i>Actuator Specification</i>	
Type	4-RW Pyramid
Max Torque [Nm]	0.05
Max Rate [RPM]	5000
J_w [kg·m ²]	4.2×10^{-4}
<i>Controller Parameters</i>	
$K_{px} / K_{py} / K_{pz}$	1.05 / 1.1 / 1.05
$K_{dx} / K_{dy} / K_{dz}$	4.2 / 4.4 / 4.2

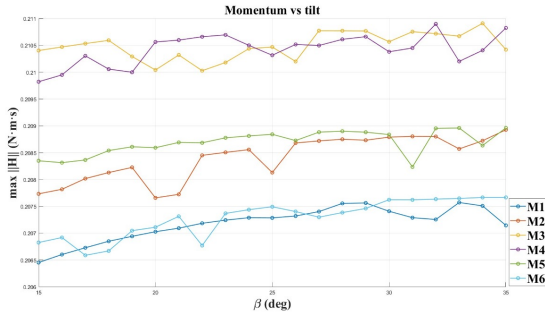


Fig. 11. Variation of RW's array angular velocity with respect to the tilt-angle, β is observed in Figure 4.

TABLE III
COMPREHENSIVE DATA OF ATTITUDE CONTROL SYSTEM'S PERFORMANCE

No.	h_{rw} [Nms]	Tilt-angle (deg)	δ_e [rad]
1	27.3234	27	1.0281e-11
2	27.1290	27	9.9648e-13
3	15.1360	31	1.0064e-13
4	16.9538	25	2.5265e-14
5	24.0292	31	7.9838e-11
6	25.9572	30	8.0067e-10

simulation parameters for the baseline configuration are provided in Table IV, including the gravity gradient disturbance model applied during the maneuvers.

The controller retained non-oscillatory convergence and negligible steady-state errors, as shown in Fig. 12, where the Euler angles smoothly converge to the desired attitude $[-30^\circ, 20^\circ, 15^\circ]$ within approximately 50 seconds. Per-wheel torque peaks stayed beneath hardware limits with no sustained clamping.

Figure 13 presents the norm of the RWs array angular momentum over the course of the maneuver. The momentum builds during the initial acceleration phase, reaches a peak near the midpoint of the slew, and then decays as the spacecraft

TABLE IV
EXTENDED SIMULATION DATA

Statement	Amount
Initial condition	$[\phi_0, \theta_0, \psi_0] = [0, 0, 0]$ deg
RWs incidence angle	$\beta = 28.5^\circ$
Desired attitude	$[\phi_d, \theta_d, \psi_d] = [-40, 20, 30]$ deg
Gravity gradient disturbance	$T_{dx} = \frac{3\mu}{2r_0^3}(I_z - I_y)\sin(2\phi)\cos^2(\theta)$ $T_{dy} = \frac{3\mu}{2r_0^3}(I_z - I_x)\sin(2\theta)\cos(\phi)$ $T_{dz} = \frac{3\mu}{2r_0^3}(I_x - I_y)\sin(2\theta)\sin(\phi)$

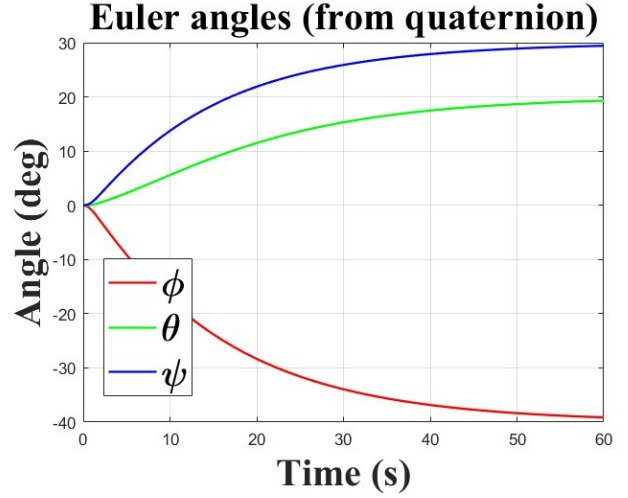


Fig. 12. Euler angles time response showing smooth convergence to the desired attitude over a 50-second maneuver.

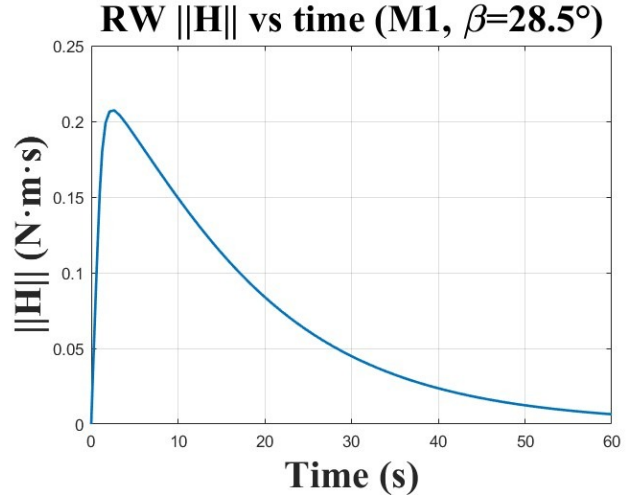


Fig. 13. Norm of the RWs angular momentum showing buildup during acceleration and decay during approach to target attitude, with peak values remaining within safe margins.

approaches the target attitude. The final momentum remains well below saturation limits, validating $\beta = 28.5^\circ$ as a practical, single-setting compromise when a mission requires a common geometry for varied slews.

Deviations of $\pm 3\text{--}5^\circ$ around 28.5° produced only gradual changes in peak momentum, indicating a forgiving design space; this is useful for mechanical tolerance and integration constraints. Under the injected sinusoidal disturbance mapped via $R(q)$, the closed loop preserved the trends above with small residuals that decay with the dominant time constant, demonstrating adequate robustness.

F. Discussion and Implications

The results support a clear design guideline: select a pyramidal tilt near $28\text{--}29^\circ$ when the objective is to minimize wheel-momentum accumulation across a representative slew set without sacrificing pointing accuracy. Missions with tightly constrained power or thermal budgets can leverage the allocator's null space to further reduce instantaneous power or to balance wheel usage while keeping the same β . If a specific slew dominates mission duty-cycle, a maneuver-specific β^* can yield additional momentum margin at the cost of re-machining or re-mounting the RWA.

G. Baseline at $\beta = 28.5^\circ$

Fixing β at 28.5° and repeating the six slews confirmed that momentum growth remains modest and decays post-slew, with low desaturation risk over the simulated horizon. The controller retained non-oscillatory convergence and negligible steady-state errors, and per-wheel torque peaks stayed beneath hardware limits with no sustained clamping. This validates $\beta = 28.5^\circ$ as a practical, single-setting compromise when a mission requires a common geometry for varied slews.

H. Sensitivity and Robustness

Deviations of $\pm 3\text{--}5^\circ$ around 28.5° produced only gradual changes in peak momentum, indicating a forgiving design space; this is useful for mechanical tolerance and integration constraints. Under the injected sinusoidal disturbance mapped via $R(q)$, the closed loop preserved the trends above with small residuals that decay with the dominant time constant, demonstrating adequate robustness.

I. Discussion and Implications

The results support a clear design guideline: select a pyramidal tilt near $28\text{--}29^\circ$ when the objective is to minimize wheel-momentum accumulation across a representative slew set without sacrificing pointing accuracy. Missions with tightly constrained power or thermal budgets can leverage the allocator's null space to further reduce instantaneous power or to balance wheel usage while keeping the same β . If a specific slew dominates mission duty-cycle, a maneuver-specific β^* can yield additional momentum margin at the cost of re-machining or re-mounting the RWA.

This paper presented a complete quaternion-PD attitude control framework for a rigid spacecraft equipped with a four-wheel pyramidal reaction-wheel array, implemented and validated in simulation. The system model included rigid-body rotational dynamics, quaternion kinematics, actuator allocation through the array geometry, and inertial-to-body disturbance mapping, enabling reproducible evaluation of large-angle slews under bounded disturbances. The closed-loop responses achieved fast, non-oscillatory convergence to the commanded attitude with negligible steady-state error, while keeping per-wheel torques within rated limits and avoiding sustained saturation. Momentum evolution remained modest and decayed post-slew, indicating a low desaturation burden for the considered maneuvers.

A geometry sweep confirmed that the pyramidal tilt angle has a first-order impact on momentum usage: across six representative rest-to-rest slews, the wheel-momentum norm exhibited a broad minimum for in the $25\text{--}33^\circ$ range, with an aggregate optimum near 28.5° . Fixing at this value delivered a practical trade-off—reduced momentum build-up without compromising pointing accuracy or torque authority—making it a suitable default when a single mechanical configuration must serve diverse slews. These findings align with prior reports on momentum envelopes and allocation trade-offs for redundant arrays, while extending them with an end-to-end quaternion-error realization and implementation-level verification.

Future work will integrate null-space biasing for power and thermal objectives, incorporate actuator and sensor non-idealities (friction, delays, misalignments, quantization), and evaluate momentum-management strategies that combine magnetorquers or thrusters for scheduled unloading. Hardware-in-the-loop tests and on-orbit disturbance characterization will further quantify robustness and refine gain/geometry co-design for mission-specific duty cycles.

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